

6.1 Coding

Question 1:

We had you define the acceleration due to gravity as a field in a structure that you had to pass as an input argument to several functions. Instead, we could have had you type the value for the constant, 3.72 m/s^2 , directly in those functions. Do you believe there is an advantage to how we had you do it? Explain. Would you have done it differently? Explain why or why not.

Instead of typing the value $g = 3.72 \text{ m/s}^2$ directly in every function, defining the gravity as a field in the structure that we had to pass as an input argument to several functions had an advantage because this made the code reusable, meaning that if I wanted to use a different planet or I didn't want to use that value anymore, I wouldn't have to change it in every code. I could just change it in one area and be able to use the same code.

Question 2:

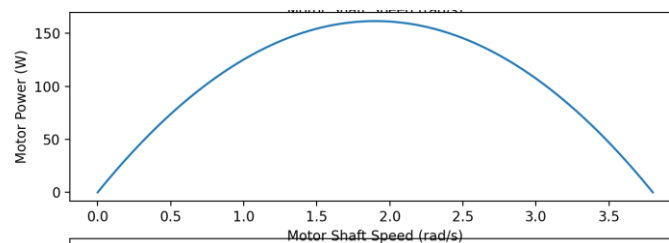
What happens if you try to call `F_gravity` using a terrain slope of 115 degrees? Is this desirable behavior? Explain why you think this

If I were to try and call `F_gravity` using a terrain slope of 115 degrees, it would pause the function and raise an exception saying that the terrain slope should be between -75 and 75 degrees. This is desirable behavior because it ensures the rover will not go outside the range of terrain it was designed for.

6.2 Motor and Speed Reducer Behavior

Question 3:

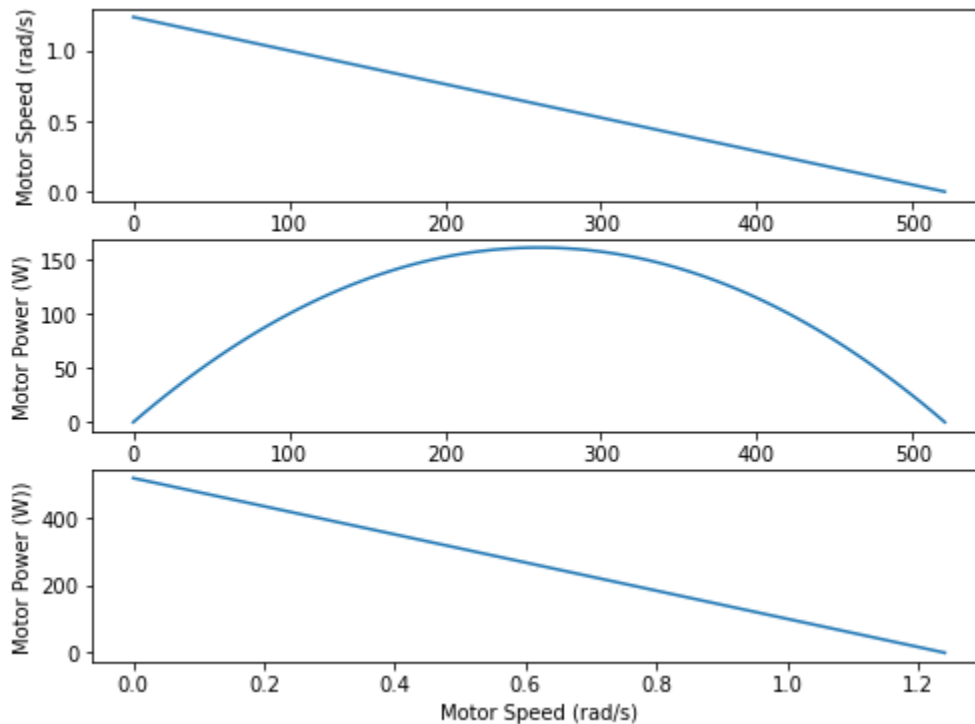
What is the maximum power output by a single rover motor? At what motor shaft speed does this occur? Provide graphs or other data to support your answer.



The maximum power output of the rover motor is approximately 160 Watts. This occurs at a motor shaft speed of about 2 rad/sec. The maximum power and corresponding motor shaft speed can be identified from the peak of the Motor Power versus Motor Shaft Speed graph.

Question 4:

What impact does the speed reducer have on the power output of the drive system? Again, provide any graphs or supporting data.



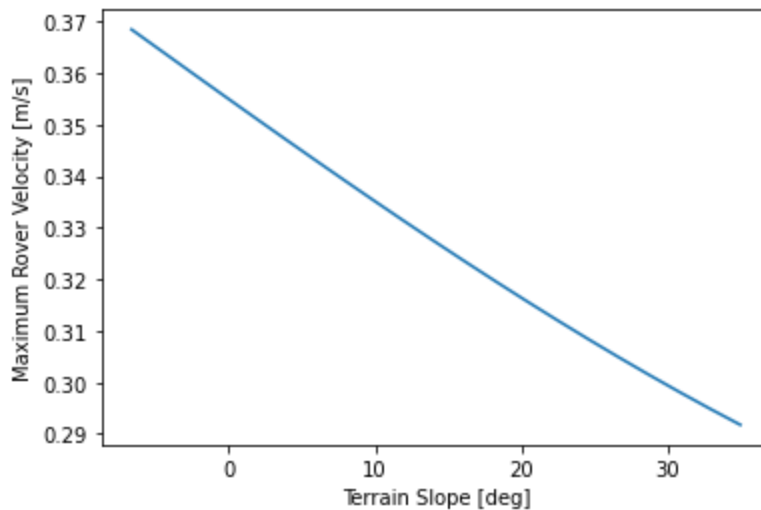
The speed reducer does not have any effect on the maximum power output of the drive system, the peak power remains the same at around 160 Watts before and after the reducer. What changes is the data point at which the peak occurs. At the motor shaft, the peak power happens at around 2 rad/sec, and after the reducer, the same peak power occurs at a lower speed of 0.63 rad/sec with a higher torque 260 Nm. This makes sense physically because different gear ratios can trade speed for torque without adding or removing energy so power stays constant through the reducer even though the torque and angular velocity would change.

6.3 Rover Behavior

Question 5:

Examine the graph you generated using `analysis_terrain_slope.py`. (Provide the graph in your response for reference.) Explain the trend you observe. Does it make sense physically? Why or why not? Please be precise. For example, if the graph appears linear or non-linear, can you explain why it should be the way you observed? Refer back to the rover model and how slope impacts rover behavior.

The maximum velocity decreases as the slope increases, which makes sense physically. On a downhill slope, the gravity's component that goes along the direction of motion adds to the driving force, allowing the rover to reach a higher speed before the motor's torque output drops

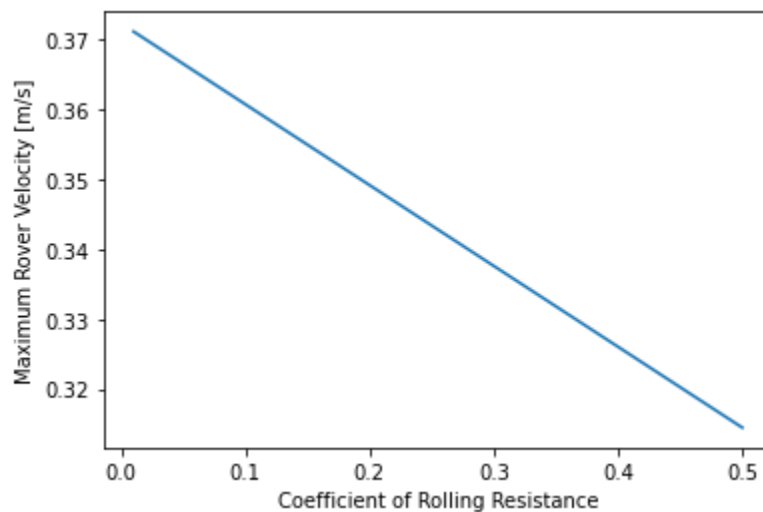


enough to balance the net force. As the slope becomes positive, gravity will oppose motion, so the rover has to settle at a lower shaft speed with a higher torque in order to reach equilibrium.

The curve is close to a straight line but it bends slightly. This traces back to the gravitational force components, which depends on $\sin(\text{slope})$. Since $\sin$ is a nonlinear function, the slope would become steeper and provide visual curvature.

Question 6:

Examine the graph you generated using `analysis_rolling_resistance.py`. (Provide the graph in your response for reference.) Explain the trend you observe. Does it make sense physically? Why or why not? Please be precise. For example, if the graph appears linear or non-linear, can you explain why it should be the way you observed? Refer back to the rover model and how the coefficient of rolling resistance impacts rover behavior.

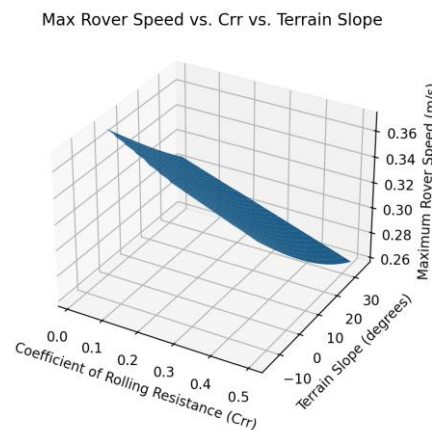


The maximum velocity decreases as the coefficient of rolling resistance increases, which makes sense physically. On a downhill slope, there would be a higher rolling resistance and thus a higher C_{rr} , which would make for the rover harder to reach a higher speed.

The relationship between maximum velocity and coefficient of rolling resistance has no noticeable curvature in comparison to the relationship between maximum velocity and terrain slope. This is because the rolling resistance formula, $F_{rr} = C_{rr} \cdot F_n$ is a constant that does not depend on speed. In order to correct it for the fact that rolling resistance should be zero when the rover is at rest, and should always oppose the direction of motion, the $\text{erf}(40 \cdot v_{\text{rover}})$ term is used to correct it and make F_{rr} a velocity dependent argument. At the velocities reached in this graph ranges around 0.32 to 0.37 m/s, the $\text{erf}(40 \cdot v_{\text{rover}})$ has already saturated to about 1, so it behaves as a constant. With erf fixed at 1, this provides the linear relationship between C_{rr} and the maximum velocity.

Question 7:

Examine the surface plot you generated using `analysis_combined_terrain.py`. (Provide the graph in your response for reference.) What does this graph tell you about the physical conditions under which it is appropriate to operate the rover? Based on what you observe, which factor, terrain slope or coefficient of rolling resistance, is the dominant consideration in how fast the rover can travel? Please explain your reasoning.



The surface plot generated shows how the rover's max velocity changes with the terrain's slope and coefficient of rolling resistance. The rover operates better on surfaces with lower rolling resistance and less steep slopes. As the terrain slope increases, the gravitational force also increases, which requires more driving force from the motors and the velocity decreases. The rover's performance is also decreased with increasing the coefficient of rolling resistance. Based on the plot, the terrain slope seems to be the dominant because the changing in slope changes the rover's maximum velocity more than the coefficient of rolling resistance. Therefore, it is important to operate on more level terrains to be able to maintain higher speeds.